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The roles of blur and eye convergence in distance estimation in larval zebrafish

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ABSTRACT

Animals use an array of visual cues to gauge distance, and their underlying neural mechanisms remain largely unknown. Zebrafish larvae execute different hunting behaviors depending on distance to the prey, providing a simple model system in which to study this process. To identify distance cues, we presented equivalent prey stimuli at increasing distance and recorded hunting behaviors. We found that the initial convergence angle was lower for more distant prey, suggesting that they are able to gauge distance to the prey via monocular cues. We investigated blur as a possible monocular cue, and found that by artificially blurring the stimulus, we were able to reduce initial convergence and strike probability. This implicates blur as a distance cue in zebrafish prey capture, adds to our knowledge of how larvae are able to visually target and accurately capture prey.

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zebrafish; visual; hunting; strike; eye convergence; blur

Introduction

Zebrafish larvae are able to use their relatively simple visual system to identify, hunt, and capture small moving prey objects, such as paramecia. Hunting begins with eye convergence and an orienting turn towards the prey, and ends with a capture strike when the prey is close enough, in which the larvae suck the prey into their mouths (Borla *et al.*, 2002; Budick & O'Malley, 2000; Hernández *et al.*, 2002; Mearns *et al.*, 2020). The degree of eye convergence is also distance dependent; binocular convergence angle is larger for more proximal prey (Patterson *et al.*, 2013). The relationship between eye convergence and the decision to strike remains unclear.

All phases of hunting behavior can be studied in head-fixed larvae and triggered with artificial stimuli (Bianco *et al.*, 2011; Khan *et al.*, 2024), which has allowed the identification of the optimal stimuli for hunting initiation and the strike (Khan *et al.*, 2023; Semmelhack *et al.*, 2014). Small moving UV dots are an ideal prey stimulus, and stimulus intensity and contrast are important in the decision to strike (Khan *et al.*, 2023). However, other visual cues may also play a role in distance estimation.

Here, we investigated the behavioral responses to prey stimuli presented at different distances. We found that the convergence angle triggered by more distant prey was smaller, despite the size, speed, and intensity of the stimulus being equivalent. We hypothesized that greater stimulus blur

at the longer distance could account for the difference. Indeed, we found that artificially adding blur that would correspond to a more distant prey caused a smaller convergences, as well as a lower chance of strikes, implicating blur as an important distance cue in hunting behavior.

Results

To investigate the visual cues for distance estimation, we presented prey on a screen two different distances from the larvae and recorded hunting initiation and strikes (head elevation (Khan *et al.*, 2023, 2024)). We first positioned the larva 3 mm from the screen, then moved the larva to 8 mm away from the screen (Figure 1A). For the 3 mm distance, we used prey stimulus parameters that robustly evoke hunting (Khan *et al.*, 2023); the dot was a UV-bright disk 4° in diameter, with a power of 5 nW and speed of 60°/s. We then moved the larva to 8 mm from the screen, and presented the same prey stimulus from the larva's perspective, increasing the size and speed so that they would still be 4° in diameter and 60°/s (Figure 1B). Using a power meter to measure the light at 3 and 8 mm (Khan *et al.*, 2024), we determined that the additional 5 mm distance caused a decrease in intensity of 10.7%, and we increased the stimulus intensity to compensate for the additional light scattering over the 8 mm distance. To quantify hunting initiation in response to these stimuli, we used binocular eye

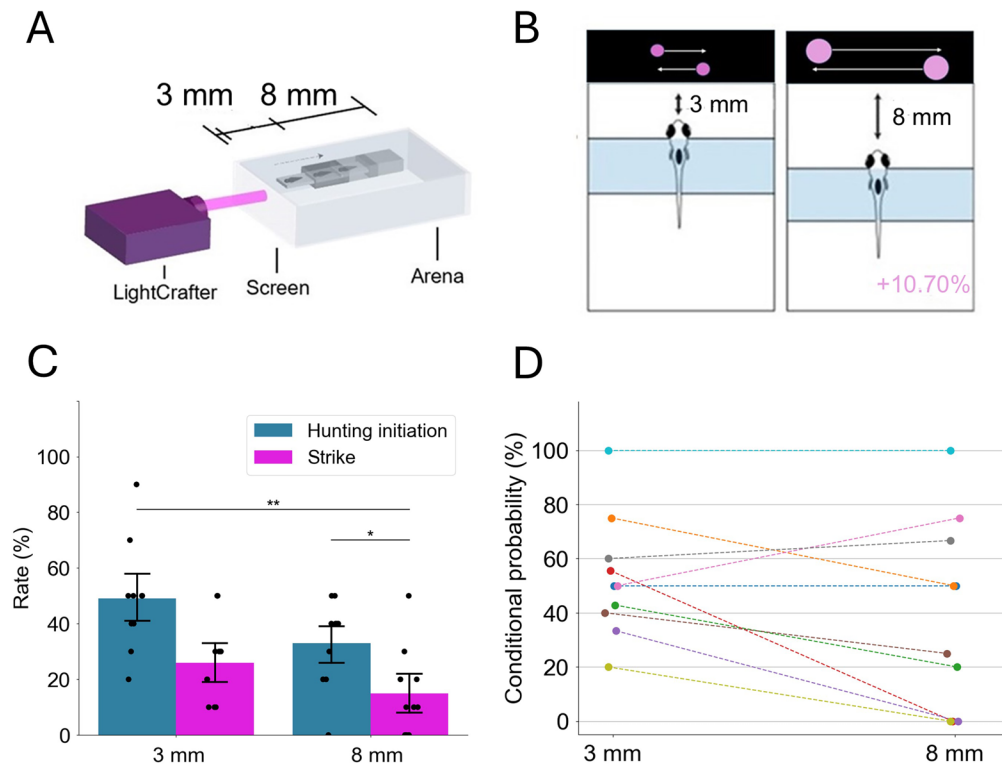


Figure 1. Behavioral responses to prey at 3 and 8 mm. A: Schematic of the behavioral setup showing movement of the larva away from the screen. B: the UV prey stimulus presented at 3 mm (4° dot moving at $20^\circ/\text{second}$) and 8 mm (same size and speed to the larva, intensity increased by 10.7% to compensate for distance). C: Hunting initiation and strike rates at 3 mm and 8 mm ($n=10$ larvae). Error bars represent SD. Black dots represent individual larvae. *, $p < 0.05$, **, $p < 0.01$, Mann-Whitney U test. D: Conditional probability of a strike given eye convergence at 3 mm and 8 mm for the same dataset. Each color represents one larva.

convergence, setting the threshold for each fish based on the position of the second peak in the bimodal convergence angle distribution (Antinucci *et al.*, 2019). Strikes were identified as upward motions of the head recorded on the side camera (Khan *et al.*, 2024).

Each trial was classified as a no response, hunting initiation only (eye convergence above the threshold but no strike) or strike (eye convergence and head elevation) trial. We calculated hunting initiation and strike rates, as a percentage of trials containing the relevant behavior, for each larva in response to each stimulus. We found that despite the equivalent size, speed, and intensity, the 8 mm stimulus triggered slightly fewer hunting only trials (Figure 1C, blue bars) and fewer trials with strikes (Figure 1C, magenta bars) but the differences in the numbers of hunting only and strike trials at 3 mm vs 8 mm were not significant. When we calculated the conditional probability of a strike occurring given hunting initiation, the conditional probability of a strike was lower at 8 mm than at 3 mm, but not significantly so (Figure 1D).

We next analysed eye convergence angles in this dataset, for the two distances and for strike and hunting initiation only trials. Larvae often have an initial convergent saccade at hunting onset (Figure 2A and B, blue dashed lines), which may be followed by additional and additional convergence later, particularly at strike onset (Figure 2B, magenta dashed lines). The initial convergence is likely driven by monocular cues, since before the eyes converge, the binocular zone is small, and hunting initiation usually occurs when the prey is lateral and can only be seen by one eye (Bianco *et al.*, 2011).

However, later convergent saccades could be driven by binocular cues. We defined the initial convergence as the angle 100 ms after hunting onset, and plotted this value for the two distances. The initial convergence angles ranged from 20° – 50° , and were significantly larger for the 3 mm condition (Figure 2C). This suggests that larvae are able to read out the distance to the screen and converge appropriately, in the absence of size, speed, and intensity differences and binocular distance cues.

We next asked whether there was a difference in initial convergence angle during hunting initiation only vs. strike trials. We found that for both distances, the initial convergence angle was greater in strike trials than in hunting initiation only trials (Figure 2D). One possibility is that for strike trials, the hunting onset might occur when the prey is closer to the center of the screen (0°), and central prey will be closer to the larva due to our use of a flat screen. However, we found that hunting initiation typically occurs around -30° and $+30^\circ$, regardless of whether the trial will have a strike (Figure 2E and F). Taken together, these results indicate that at the longer distance, there is some difference in the stimulus that induces less initial convergence on average. However, for the smaller number of 8 mm trials where the initial convergence angle was large, a strike tended to follow. This suggests that the distance estimation and decision to strike may already have been determined at hunting onset, and therefore it is likely that monocular cues are involved.

One candidate monocular cue could be blur, since a more distant prey will be out of focus and thus blurred to the

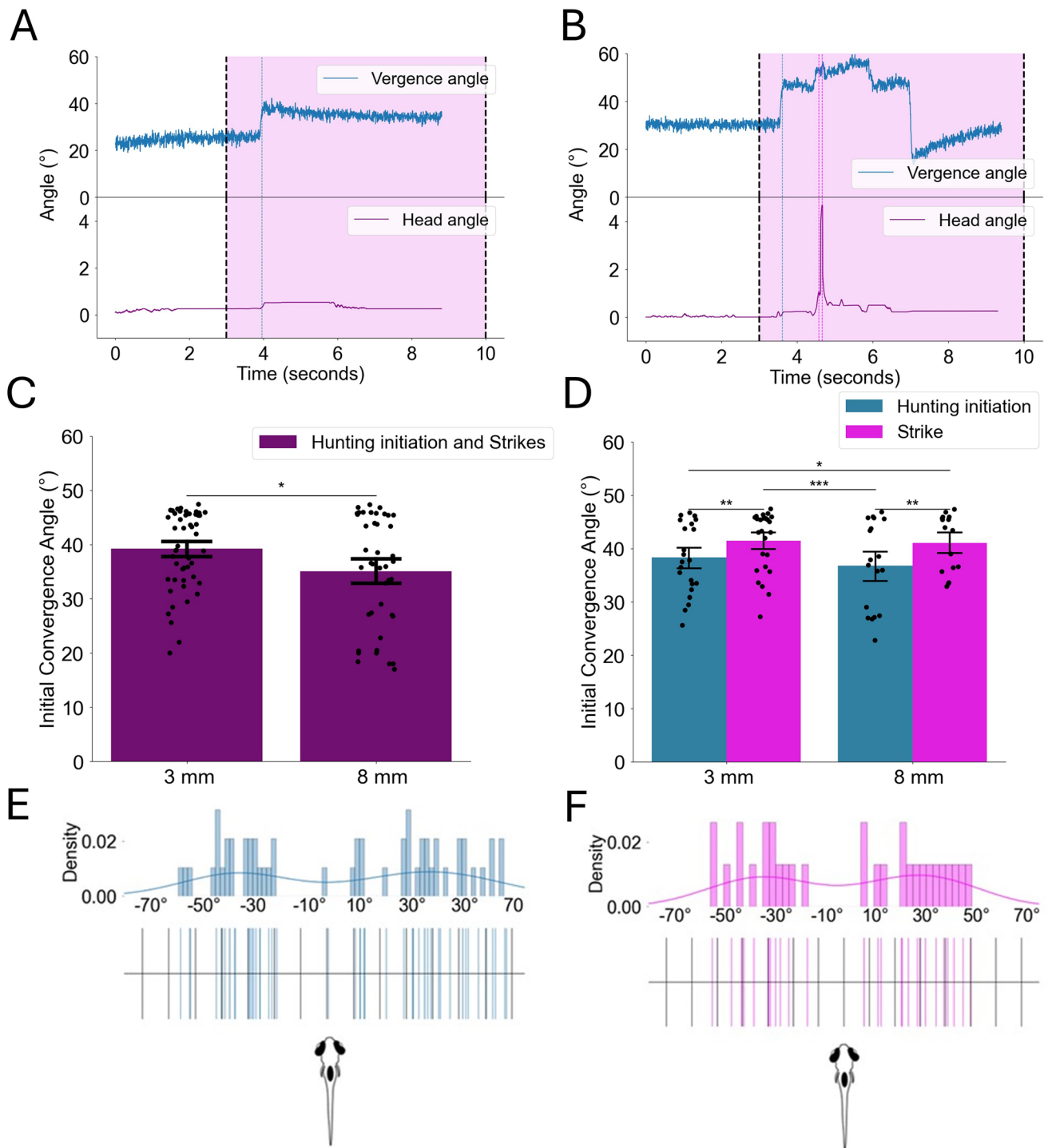


Figure 2. Convergence is greater in at 3 mm and in strike trials. **A.** Binocular vergence angle and head angle over time in response to prey stimuli (pink shading) at 3 mm in a hunting initiation only trial. Blue dotted line represents hunting onset. **B.** Vergence and head angle in the same fish during a strike trial. Magenta dotted lines represent strikes. **C.** Initial convergence angle 100ms after hunting onset for all trials at 3 mm and 8 mm ($n=10$ larvae). Black dots represent initial convergence angle in individual trials. **D.** Initial convergence angle in hunting initiation only and strike trials at the two distances. Black dots represent individual trials. *, $p < 0.05$, **, $p < 0.01$, ***, $p < 0.001$, Mann-Whitney U test. **E.** Horizontal position of the prey at hunting onset in hunting initiation only trials (blue lines). **F.** Horizontal position of the prey at hunting onset in strike trials (magenta lines).

larva. We set out to test the effect of blur on the hunting response by adding a gaussian blur of up to 2.5 standard deviations to our 4° UV prey dot (Figure 3A). We calculated that a prey distance of 8 mm would correspond to a blur of slightly more than 1.5 SD (methods). With the screen 3 mm away, we found that increasing blur decreased the strike rate as well as the hunting rate (Figure 3B). The increasing blur

also decreased the conditional probability of strikes given hunting initiation (Figure 3C). We then plotted the initial convergence angles for different amounts of blur, and found that increasing stimulus blur significantly reduced the initial convergence angle (Figure 3D). This suggests that blur is a monocular cue for distance estimation, and that more distant and blurred stimuli trigger a smaller degree of eye

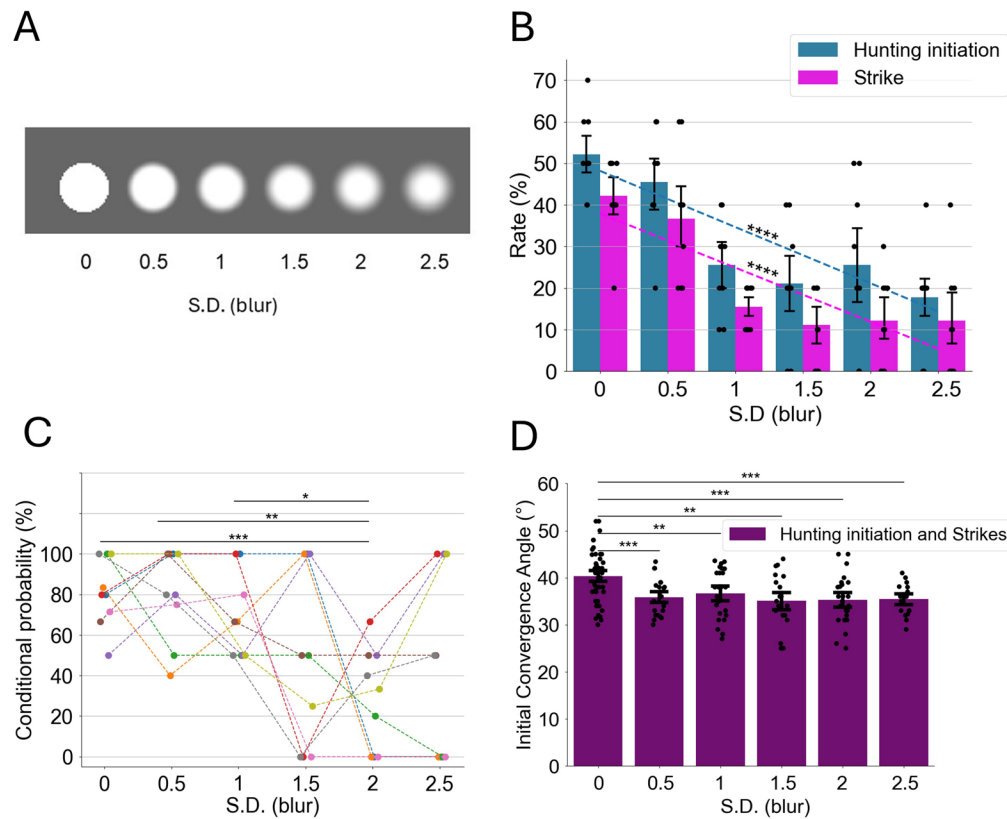


Figure 3. Blur decreases strike probability and causes a weaker initial convergence. A. Gaussian blurring of the prey dot with different standard deviations; blur the blur expected from a distance of 8 mm is slightly more than 1.5 SD (methods). B. Hunting and strike rates for prey at presented at 3 mm with different degrees of blur. Black dots represent individual larvae. Dashed lines represent linear regression best fit lines, ****, $p < 0.0001$, Mann-Whitney U test. ($n = 10$ larvae). C. Conditional probability of strikes given eye convergence at different degrees of blur. Each color represents one larva. *, $p < 0.05$, **, $p < 0.01$, ***, $p < 0.001$, Mann-Whitney U test. D. Initial convergence angle at 3 mm for prey stimulus with varying degrees of blur. Black dots represent individual trials.

convergence during hunting, and are less likely to evoke strikes.

Discussion

Zebrafish have become a popular system for studying visual behaviors and neural circuits, as they are highly visual and use their visual system to estimate distance to prey (Patterson *et al.*, 2013). They also have mobile eyes, and execute a binocular convergence at the onset of prey capture (Budick & O'Malley, 2000), with the extent of convergence being related to prey distance (Patterson *et al.*, 2013). Eye convergence and distance estimation thus seem to be related, but the exact nature of the relationship remains unclear. Here, we set out to investigate how zebrafish larvae judge distance, using eye convergence and strike behavior as the readouts. We found that a greater distance to the prey did decrease the amount of convergence in response to the prey stimulus, and that trials with less initial convergence are less likely to include strikes. We hypothesized that stimulus blur could be one effect of the greater distance, and indeed found that blurring the prey stimulus causes a decrease in strike probability and initial convergence angle.

Blur is an important visual depth cue in humans (Held *et al.*, 2012) and can also drive eye convergence (Horwood & Riddell, 2008). Our results indicate that blur also plays a role in these processes in zebrafish larvae. But, as in humans, it is

difficult to disentangle vergence from distance estimation. One possibility is that blur, due to greater distance or our artificial blurring of the stimulus, causes a lower degree of initial convergence, and that in turn makes a strike less likely. Binocular vision is thought to be important for the final phase of prey capture in zebrafish larvae (Mearns *et al.*, 2024), so perhaps having a larger binocular zone as the result of full convergence could facilitate strikes. It is also possible that larvae are using convergence angle to triangulate the distance to prey (Read, 2023), in which case greater convergence could lead to the perception that the prey is close and therefore make strikes more likely. We find that regardless of distance, trials with a higher degree of convergence are more likely to include strikes (Figure 2D), indicating that these processes are intimately linked. Further experiments will be required to distinguish between the different mechanisms.

We previously found that prey stimulus intensity and contrast affect the proportion of hunting and strike trials, and thus are likely to play roles in distance estimation (Khan *et al.*, 2023). Here, we have shown that blur is another important visual cue in prey capture, modulating the degree of binocular convergence and the likelihood of strikes. However, whether blur directly effects distance estimation and thus strikes or whether the effect is due to the reduced convergence remains to be determined. The present work adds to the stimulus parameters larvae use in deciding whether to hunt or strike in response to a prey object. It is

likely that larvae rely on multiple cues to estimate distance, and it will be interesting to ask in the future how these parameters interact with each other.

Methods

Animals

All procedures were approved by the animal ethics committee of HKUST. Zebrafish carrying mutations in the *mitfa* allele (nacre) were used for all experiments, and were housed under a standard 14:10 light cycle. Larvae were kept in Danieau's solution, and fed with paramecia (*Paramecium multimicronucleatum*, Carolina Biological Supply Company) at 5 and 7 dpf.

One day before behavioral experiments, 6, 7 or 8 dpf larvae were embedded in 2% low melt agarose on Rinzl Plastic Coverslips (Delta Microscopies), and the eyes and tail were freed from agarose with a razor blade, as previously described (Khan *et al.*, 2023, 2024).

Behavioral assay

The behavioral tests were conducted in a chamber where one side consisted of a Teflon fabric screen (Khan *et al.*, 2023). A camera on the side of the larva (BFS-U3-04S2M-CS USB 3.1 FLIR Blackfly S, Monochrome Camera) fitted with a right angle zoom lens (VZM 450i Right Angle Zoom Imaging) was used to record head elevation, and a second camera (Photon Focus, MV1-D1312E-160-CL-12), fitted with an objective lens (Sigma 105mm), under the arena was used to record tail and eye movements. Both the side and bottom camera recorded at a framerate of 200 frames/second.

At the start of the experiment, the coverslip on which the larva was embedded was placed into a 3D printed coverslip holder, which was attached to an XYZ stage (Dovetail slide stage, HK Mechanics). It was then lowered into the arena filled with Danieau's solution. The Light Crafter (LCr) visual stimulator was placed in front of the screen at a distance of 13.5 cm.

Spatial visual stimulator

We used a DLP LightCrafter E4500 MKII from EKB Technologies to project UV prey stimuli, using a UV LED with a peak at 385 nm.

Visual stimuli

Visual stimuli were designed in Python using PsychoPy (Peirce *et al.*, 2019). The prey stimulus was a 4° diameter UV (385 nm) dot on a black background, sweeping at a speed of 60°/second horizontally from -70° degree to 70°, with an elevation of 27° degrees. The interstimulus interval was 1 min, with a recording period of 10 s. Stimulus order was randomized.

To measure the attenuation of the stimulus in water, we projected a 50° diameter dot on the screen and measured

the intensity with a power meter (Thorlabs, PM100USB). We submerged the power meter sensor (Thorlabs, S130VC 200 nm – 1100 nm) in the behavioural chamber at the position of the fish, and calculated the intensity of a 4° prey stimulus assuming power per area is constant. To change the stimulus intensity to compensate for the 8 mm distance, the LED power was adjusted by turning the knob on the driver attached to the UV, LED, and the power meter was used measure the resulting stimulus.

Eye tracking and hunting initiation

Eye tracking was performed as previously described (Khan *et al.*, 2024), with a custom python code using the OpenCV python library. The binocular eye angle distribution was fitted to a kernel density estimation using a Gaussian Kernel. The distributions were predominately bimodal, and the threshold for hunting initiation for each fish was defined as one standard deviation below the second peak in the bimodal distribution (Antinucci *et al.*, 2019). In cases where the vergence angle distribution was not bimodal, we used a fixed threshold of 50°.

Strike classification

Strikes were identified from side camera data as previously described (Khan *et al.*, 2023), using the Farneback optical dense flow algorithm (Farneback, 2003) to identify pixels with upward motion. And defining strike frames as those where the mode of the moving pixels was upward.

Stimulus blur

The prey stimulus was blurred by applying a gaussian filter in psychopy, with standard deviation of 0–2.5. We calculated the expected blur of a prey object 8 mm away as follows: we used a pupil diameter of 250 μm (Wang *et al.*, 2020) and focal length of 252 μm (Young *et al.*, 2018) to calculate the blur radius as previously described (Strasburger *et al.*, 2018). The expected blur was 1.68°. We then determined the width of the blurred ring around each of our blur stimuli, and found that our stimulus with blur SD = 1.5 had a blur radius of 1.55°, while the stimulus with SD of 2 had a blur radius of 2.70°, indicating that the 8 mm distance blur should be between these two stimuli and close to SD = 1.5.

Statistics

Statistical tests were performed using the SciPy library in Python. All the data was first tested to check if they are from a normal distribution or not using the Kolmogorov Smirnov test. If the *p* values of this test was significant it was concluded that the data don't follow a normal distribution. Since all our data didn't follow the normal distribution, then for the comparison across different groups the Kruskal-Wallis test (non-parametric Anova test) was conducted. If the *p* values of this test was significant, then we used the Mann-Whitney U (non-parametric test) for doing

the pairwise comparison across these groups to find the significantly different pairs.

P values ≤ 0.0001 was denoted by “****”

P values ≤ 0.001 was denoted by “***”

P values ≤ 0.01 was denoted by “**”

P values ≤ 0.05 was denoted by “*”

For the linear regression we used the model from stats-model in Python to find the best fit line.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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References

- Antinucci, P., Folgueira, M., & Bianco, I. H. (2019). Pretectal neurons control hunting behaviour. *eLife*, 8, e48114. <https://doi.org/10.7554/eLife.48114>
- Bianco, I. H., Kampff, A. R., & Engert, F. (2011). Prey capture behavior evoked by simple visual stimuli in larval zebrafish. *Frontiers in Systems Neuroscience*, 5(December 2011), 101. <https://doi.org/10.3389/fnsys.2011.00101>
- Borla, M. A., Palecek, B., Budick, S., & O'Malley, D. M. (2002). Prey capture by larval zebrafish: Evidence for fine axial motor control. *Brain, Behavior and Evolution*, 60(4), 207–229. <https://doi.org/10.1159/000066699>
- Budick, S. A., & O'Malley, D. M. (2000). Locomotor repertoire of the larval zebrafish: Swimming, turning and prey capture. *The Journal of Experimental Biology*, 203(Pt 17), 2565–2579. <https://doi.org/10.1242/jeb.203.17.2565>
- Farneback, G. (2003). Two-frame motion estimation based on polynomial expansion. In J. Bigun & T. Gustavsson (Eds.), *Image analysis, lecture notes in computer science* (pp. 363–370). Springer.
- Held, R. T., Cooper, E. A., & Banks, M. S. (2012). Blur and disparity are complementary cues to depth. *Current Biology: CB*, 22(5), 426–431. <https://doi.org/10.1016/j.cub.2012.01.033>
- Hernández, L. P., Barresi, M. J. F., & Devoto, S. H. (2002). Functional morphology and developmental biology of zebrafish: Reciprocal illumination from an unlikely couple. *Integrative and Comparative Biology*, 42(2), 222–231. <https://doi.org/10.1093/icb/42.2.222>
- Horwood, A. M., & Riddell, P. M. (2008). The use of cues to convergence and accommodation in naïve, uninstructed participants. *Vision Research*, 48(15), 1613–1624. <https://doi.org/10.1016/j.visres.2008.04.023>
- Khan, B., Jaesiri, O.-M., Lazarte, I. P., Li, Y., Tian, G., Zhao, P., Zhao, Y., Ho, V. D., & Semmelhack, J. L. (2023). Zebrafish larvae use stimulus intensity and contrast to estimate distance to prey. *Current Biology: CB*, 33(15), 3179–3191.e4. <https://doi.org/10.1016/j.cub.2023.06.046>
- Khan, B., Lazarte, I. P., Jaesiri, O., Zhao, P., & Semmelhack, J. L. (2024). Protocol for using UV stimuli to evoke prey capture strikes in head-fixed zebrafish larvae. *STAR Protocols*, 5(1), 102780. <https://doi.org/10.1016/j.xpro.2023.102780>
- Mearns, D. S., Donovan, J. C., Fernandes, A. M., Semmelhack, J. L., & Baier, H. (2020). Deconstructing hunting behavior reveals a tightly coupled stimulus-response loop. *Current Biology: CB*, 30(1), 54–69.e9. <https://doi.org/10.1016/j.cub.2019.11.022>
- Mearns, D. S., Hunt, S. A., Schneider, M. W., Parker, A. V., Stemmer, M., & Baier, H. (2024). Diverse prey capture strategies in teleost larvae. *eLife*, 13, RP98347.
- Patterson, B. W., Abraham, A. O., MacIver, M. A., & McLean, D. L. (2013). Visually guided gradation of prey capture movements in larval zebrafish. *The Journal of Experimental Biology*, 216(Pt 16), 3071–3083. <https://doi.org/10.1242/jeb.087742>
- Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., Kastman, E., & Lindeløv, J. K. (2019). PsychoPy2: Experiments in behavior made easy. *Behavior Research Methods*, 51(1), 195–203. <https://doi.org/10.3758/s13428-018-01193-y>
- Read, J. C. A. (2023). Stereopsis without correspondence. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 378(1869), 20210449. <https://doi.org/10.1098/rstb.2021.0449>
- Semmelhack, J. L., Donovan, J. C., Thiele, T. R., Kuehn, E., Laurell, E., & Baier, H. (2014). A dedicated visual pathway for prey detection in larval zebrafish. *eLife*, 3, e04878. <https://doi.org/10.7554/eLife.04878>
- Strasburger, H., Bach, M., & Heinrich, S. P. (2018). Blur unblurred—A mini tutorial. *i-Perception*, 9(2), 2041669518765850. <https://doi.org/10.1177/2041669518765850>
- Wang, K., Vorontsova, I., Hoshino, M., Uesugi, K., Yagi, N., Hall, J. E., Schilling, T. F., & Pierscionek, B. K. (2020). Optical development in the zebrafish eye lens. *FASEB Journal: Official Publication of the Federation of American Societies for Experimental Biology*, 34(4), 5552–5562. <https://doi.org/10.1096/fj.201902607R>
- Young, L. K., Jarrin, M., Saunter, C. D., Quinlan, R. A., & Girkin, J. M. (2018). Non-invasive in vivo quantification of the developing optical properties and graded index of the embryonic eye lens using SPIM. *Biomedical Optics Express*, 9(5), 2176–2188. <https://doi.org/10.1364/BOE.9.002176>